

Course syllabus

TEC 5143- Research in Technology

Term: Fall 2026

Course Time: Tuesday: 6:00 PM to 8:30 PM

Classroom: Klehm Hall 4431

Course Dates: Aug 24, 2026 - Dec 18, 2026

Faculty: Dr. Reachsak Ly

Email: rlly@eiu.edu (Preferred communication method)

Office: Klehm Hall 4014

Office Hours:

Monday, Wednesday and Friday (10:00 AM to 11:00 AM)

or by appointment (Virtually via Zoom or Team)

Communication Response Time: Within 2 days

Required Textbooks

Practical Research: Planning and Design, (11th Edition), Paul D. Leedy; Jeanne Ellis Ormrod Publisher ISBN: 9780133741322 ([Textbook](#))

Course description

This course introduces the means and methods of research used to define and investigate problems in technology-related fields. Topics include problem and scope definition, literature review, research methodologies, data collection, and data analysis. Students will learn the processes and tools for conducting both experimental and non-experimental research, with an emphasis on research design, analysis, interpretation, and reporting. The course also guides students in the development and writing of a research proposal.

Course objectives

Objectives of this course is to prepare students to conduct research in different technological fields. Students will be able to learn to define a research problem, to conduct effective research using library and online resources, write literature reviews and research reports, and to present their research findings in a professional manner in a group environment.

Student Learning outcome

At the end of the course the student should be able to understand the knowledge and skills required to:

1. Generate a research topic and a research problem statement and develop an empirical research question
2. Use the reference area of the library, interlibrary loan services, and computerized databases to locate relevant articles on a selected topic.
3. Execute literature reviews, Search and analyze literature to identify a gap in empirical knowledge.
4. Formulate a hypothesis statement.
5. Critical read reviews of research and critique as to:

- a. Definition of a topic or problem
 - b. Adequacy of prior reviews in the area
 - c. Selection of studies to be reviewed.
 - d. Analysis of study findings and pertinent characteristics
6. Develop a system for collecting data about the characteristics and results of previously conducted research.
 7. Analyze this collected data and interpret and report the results.
 8. Use APA standards when writing a research proposal.
 9. Develop a deliverable Research Portfolio that can be used in further research.
 10. Describe ethical issues related to empirical research and expectations for ethical behavior
 11. Describe the steps required to create an effective empirical research proposal
 12. Evaluate appropriateness of data collection and analytical methods to answer empirical research questions

Grading Structure

Content	Percentage
Discussion and participation	10%
Literature review	15%
PowerPoint presentation	15%
Quizzes	10%
Journal entries	10%
Proposal related assignment	15%
Final Proposal	10%
In-class activities	15%
Total	100%

Grading Scale:

Individual grade items and your final grade will be assessed as follows:

- A = 100- 90%
 B = 89- 80%
 C = 79-70%
 D = 69-60%
 F = less than 60%

Communication and Response Time

I encourage regular communication and engagement throughout the course.

Students may contact me primarily through email (rly@eiu.edu) or D2L Messages, which I check regularly during weekdays (working or business days). You can expect a response to your emails or messages within 24–48 hours (Monday through Friday). Emails received over the weekend or during holidays will be answered on the next business day.

Announcements and Updates: Course updates, reminders, and important announcements will be posted in the D2L News section (Activity Feed) at least once per week, or more frequently as needed to support your progress in the course. Please make it a habit to review course announcements regularly.

Professional and respectful communication is expected in all interactions in this course. When engaging through D2L discussions, emails, or virtual meetings, please follow these guidelines:

- Be courteous and respectful of others' ideas and perspectives.
- Use appropriate language.
- Write in complete sentences with proper spelling and grammar.

- Stay on topic in discussions and contribute thoughtfully to the learning community.
- Allow 24 to 48 hours for responses to emails or discussion replies.

Grading and Feedback

Assignments and assessments will be graded in a timely manner to support your learning. You can generally expect grades and feedback to be posted within one week (7 days) after the assignment due date. For major projects or written reports that require more detailed review, grading may take up to 14 days.

Feedback Types: Students will receive feedback in the form of rubrics, written comments, or summary notes in D2L, depending on the type of assignment. Feedback will highlight strengths, areas for improvement, and suggestions for future work to help you continuously improve your performance throughout the course.

Reading Assignments

Students are expected to read the textbook chapters and course slides in the relevant Module before starting each course session on the D2L.

Participation

Face to Face (F2F) courses students will need to attend class and participate in classroom discussions or activities in order to receive participation points. We will have either have activities or online based activities to show participation for the week. Partial participation points will be given in certain approved circumstances with official proof provided, (i.e., doctors note, accident report, etc.)

Journal Entries

Journal entries are not for reciting facts. It is a self-reflection on what you have learned and how you plan to use or have used the information. You are required to complete a journal entry every week for 12 weeks during the semester. See the "Journal Entry Requirements" PDF document in the Assignment Explanation Module.

Literature Reviews

Each student is asked to provide a list of three scholarly articles that are relevant to his/her own research topic interest. Students will use these articles to create their qualitative, or quantitative research proposals. This exercise is also designed to provide the opportunity for students to demonstrate their skill at completing some basic steps necessary to conduct literature reviews. See the "Literature Review Requirements - Sample" PDF document in the Assignment Explanation Module.

PowerPoint Presentations

Each student will present their Research Project in a series of two PowerPoint presentations. The first will be a 5-minute presentation based on their Research Topic, the Research Problem, Sub-problems and your Hypothesis. This will be completed on Week 7. The second will be a 10-minute presentation based on their Proposal Introduction, Problem Statement, Objectives, Preliminary Literature Review, and Methodology. This will be completed in the final class meetings. Attendance is mandatory, even if you have already presented your Research.

Quizzes

The quiz date will be announced in advance and will cover material from specified chapters in the textbook and from PowerPoint slides in the course Modules. Each quiz will be worth 25 points based of 25 questions from True and False or Multiple Choice. Each quiz will be available on D2L.

Research Project Proposal

There will be an individual project proposal that each student will write based on the instructions provided in the D2L. There will be a series of assignments based on the components required to create a proposal. Students will prepare and submit their components in an approved format and APA style in a Word document. The details of the project will be announced on D2L throughout the semester. See the "A sample proposal with comment" and "Sample Proposal Format" PDF documents in the Assignment Explanation Module.

In-class Activities

Some in-class activities and additional assignments (Library, APA and Proposal Correction) are required during the semester. You are responsible to read the description of these assignments on D2L and complete them with instructions when the appropriate Module is discussed. Some of these exercises will require submission via d2l.

Course outline

* The course schedule is subject to change throughout the semester. Revisions will be noted in class.

Week	Date	Class topics	Reading	Assignments and Quizzes
Week 1	Aug 25	Course introduction Introduction to empirical research		
Week 2	Sept 1	The Nature and Tools of Research	Chapter 1	
Week 3	Sept 8	The Problem: The Heart of the Research Process Problem Statements Review of the Related Literature Literature Analysis	Chapter 2 Chapter 3	
Week 4	Sept 15	Literature search activity Demo on Literature search Demo on Zotero Citation Plagiarism	Chapter 3	
Week 5	Sept 22	Demo on Literature search Demo on Zotero Planning your Research Project Research Questions	Chapter 4 Chapter 5	
Week 6	Sept 29	Writing the Research Proposal Example of Proposal Martini glass framework for Research exercise	Chapter 6	Quiz #1
Week 7	Oct 6	Quantitative Descriptive Research Data collection Survey Qualtrics activities		
Week 8	Oct 13	Analyzing Quantitative Data Analyzing different types of survey data Analyzing Quantitative Data Descriptive Statistics	Chapter 6	

Week 9	Oct 20	Inferential Statistics lecture	Chapter 7	
Week 10	Oct 27	Mid -Term Presentations	Chapter 9	Quiz #2
Week 11	Nov 3	Human Subject Research	Chapter 10	
Week 12	Nov 10	Human Subject Research CITY Training	Chapter 11	
Week 13	Nov 17	Online Portfolio Workshop		
Week 14		Thanksgiving		
Week 15	Dec 1	Final Presentations		
Week 16	Dec 8	No Final Exam Qualitative Research Methods Final Proposal Submission		Quiz #3

Note: This is a tentative schedule, and subject to change as necessary. The instructor reserves the right to make changes to the topics or sequence of topics, schedule, assignments, grading plan, and content of the course.

Technology Requirements:

Students must have access to a reliable high-speed internet connection, an up-to-date web browser (Chrome or Edge recommended), a webcam, a microphone, and access to Microsoft 365 (provided by EIU) and D2L (Desire2Learn).

Technical Support:

For any technical issues, please contact the EIU Help Desk at support@eiu.edu.

Academic integrity

Students are expected to maintain principles of academic integrity and conduct as defined in EIU's Code of Conduct <https://www.eiu.edu/deanofstudents/conductcode.php>. Violations will be reported to the Dean of Students. <https://www.eiu.edu/deanofstudents/>

Students with disabilities

If you are a student with a documented disability in need of accommodations to fully participate in this class, please contact the Office of Accessibility and Accommodations. All accommodations must be approved through Office of Accessibility and Accommodations. Please stop by McAfee Gym, Room 1336, or call 217-581-6583 to make an appointment.

Academic Support Center

Students who could use additional help achieving their academic goals are encouraged to contact the Academic Support Center <https://www.eiu.edu/academicsupportcenter/>. The Academic Support Center provides individual consultations, as well as assistance with time management, test taking, note taking, avoiding procrastination, setting goals, and other skills to support academic achievement. In addition, Academic Alerts may be utilized by faculty to help students who are not making academic progress or are not attending classes regularly get connected with the Academic Support Center and other important resources. To make an appointment, call 217-581-6696, or go to McAfee Gym, Room 2230.